a

SIZE ONLY (mask area) day 30 | CNN, whole image day 30 | CNN, cropped to organoid day 30 | CNN, cropped + brightened day 30 | LSTM: the whole image series: day 3 | day 10 | day 17 | day 24 | day 30

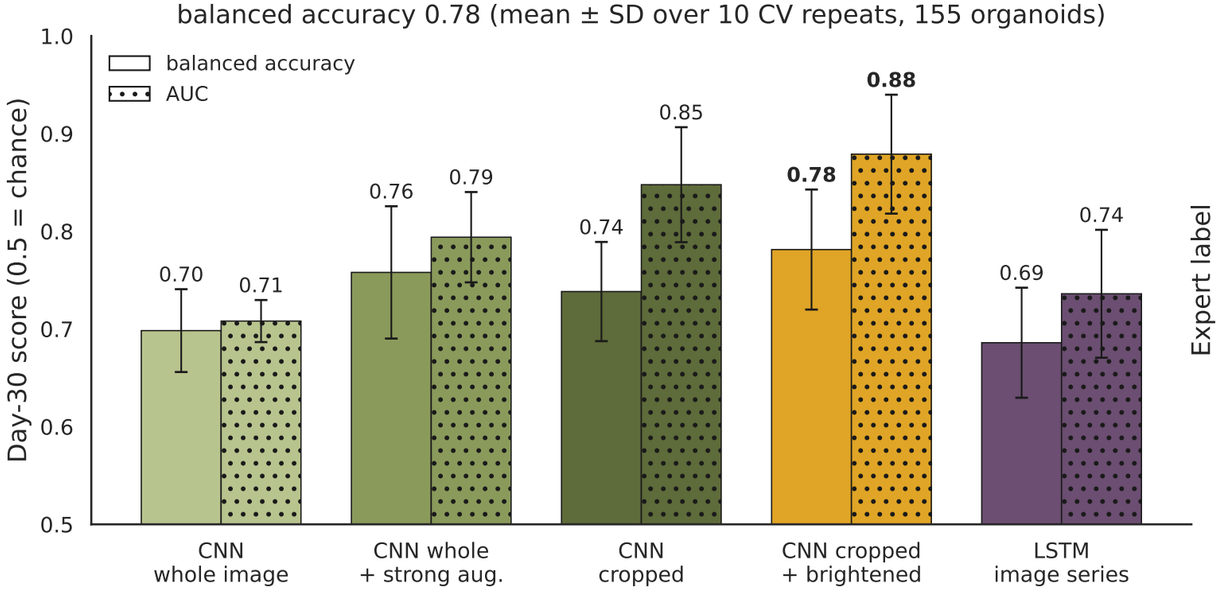
BA1 1_H6 accepted (5/5 Acc. votes) — area = 2.25 mm²

BA2 2_C3 rejected, stalled (0/5 Acc. votes) — area = 0.58 mm²

BA2 2_E12 rejected, overgrown (1/5 Acc. votes) — area = 7.44 mm²

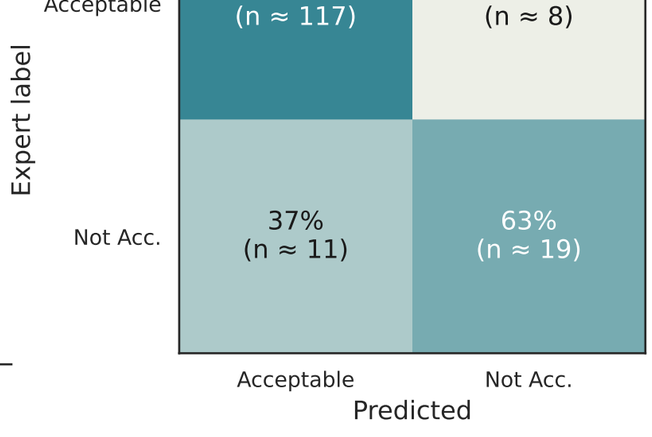
b

Day 30: the best model (CNN cropped + brightened) reaches AUC 0.88, balanced accuracy 0.78 (mean ± SD over 10 CV repeats, 155 organoids)



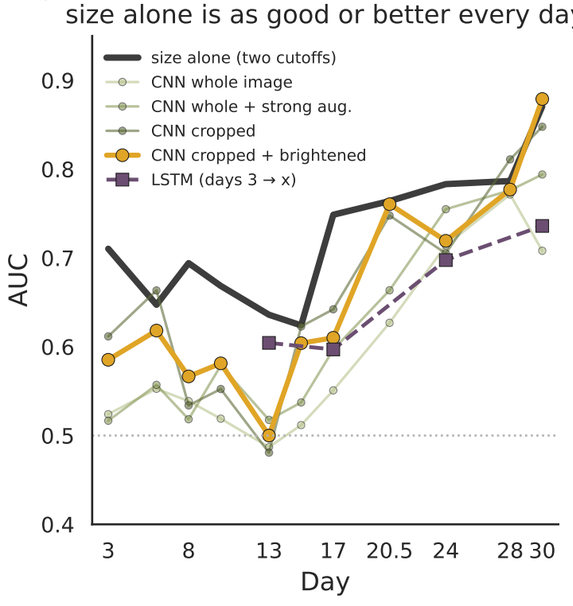
c

Best model, day 30: % of each expert class (mean organoid count per CV repeat)



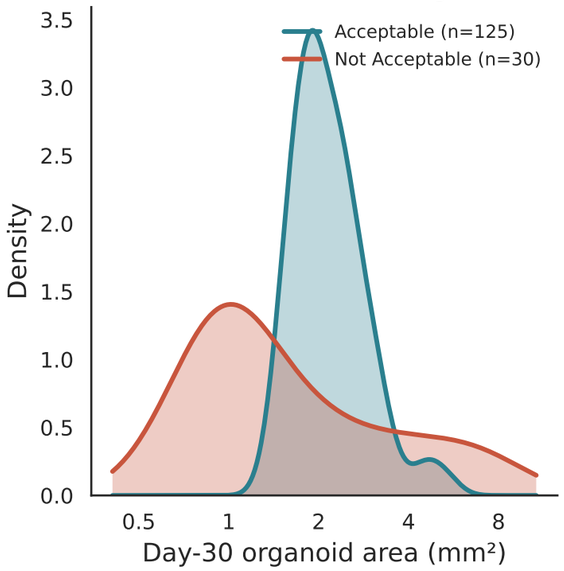
d

Image models only work from ~day 20; size alone is as good or better every day



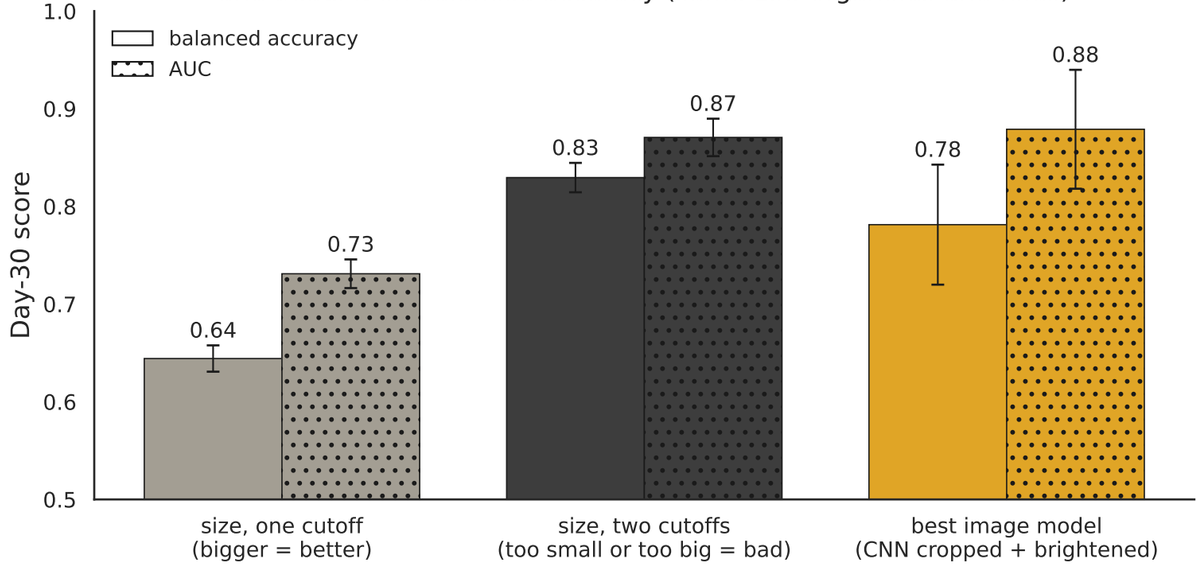
e

Not Acceptable organoids are too small or too big



f

One number (area, two cutoffs) matches the best image model on AUC and beats it on balanced accuracy (same 155 organoids and folds)



g

The image series adds nothing: LSTM ≤ size + growth rate in every window

